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Special article

Design of medical devices—A home perspective

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ABSTRACT

Health care services are moving out to the community and into the home; e-health services, remote monitoring technology and self-management are replacing hospitalization and visits to medical clinics and custom-tailored medicines are making inroads into normative treatment. These developments have great implications for the scope and design of home health care equipment.

The paper discusses the unique nature of home medical devices, from a human–environment–machine perspective, focusing on the nature of users, environment and tasks performed.

We call for increased awareness and active continuous involvement of health care personnel together with bioengineers, human factors experts, architects, designers and end users—patients and caregivers—in defining the objectives of health care devices and services at home in terms of "all family" use, integrated into the overall surroundings ("smart home"), and as part of a collaborative patient–physician disease management team.

1. Introduction

Medical devices and equipment constitute a growing sector worldwide. Modern technologies are expanding the capabilities of health care, with the resulting great economic opportunities serving as a driving force for innovation [1,2]. In parallel to the increasing specialization and sophistication of hospital equipment, a whole range of medical devices and products is moving out of hospitals and clinics into community and residual settings for use by the general public.

The escalating need for home health care devices and services is mostly associated with global demographic changes and developments in the health care system [1,3]. The graying of society, as a result of increased life expectancy and deceased birth rate, is being accompanied by a rise in chronic diseases such as diabetes, heart disease, hypertension, COPD, arthritis, osteoporosis, and depression [1]. Studies have shown that chronic diseases may be controlled and prevented by patients caring for themselves via self-care and monitoring (patient empowerment), thereby transforming the care process into a continuous collaborative relationship between patients and health care providers [1,3,4].

Developments in e-health and telemedicine have facilitated the transfer of health care services from hospitals to the community and homecare settings [3]. Advances in information communication technology (ICT), computing and remote home monitoring, electronic medical records, sensing technologies, wireless technology, virtual reality and robotics have enabled the transition from institution-centric to patient-centric treatment and care at the point of need (PON) [3–5].

The assertion that "health is a state of complete physical, psychological and social well being, not only the absence of illness," [6] combined with improved technology, has further expanded the scope and variety of home health care devices and services.

In parallel, the design profession is responding to global needs and priorities. Social design, a concept identified mostly with Papanek in his famous book *Design for the Real World* [7], expresses the designers' duty to design for the needs of society—from developed countries through to the special requirements of the aged, the poor, the disabled and the sick. This school of thought is fueling the interest and output of designers working in the area of health care devices and services.

This paper discusses the unique nature of home health care devices and services from a human–environment–machine perspective in order to highlight their special requirements and problems. Increased awareness and active continuous involvement of the medical community together with bioengineers, social workers, psychologists, designers and the end users—patients and caregivers—is needed to develop this important area.

2. Users of health care equipment and services

Users of medical equipment in hospitals and clinics are professional, experienced staff, essentially healthy and physically fit, who operate the equipment on patients. All stages—from deciding or remembering to perform the procedure, through its execution, reading the data, documenting and interpreting the results—are

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performed by trained personnel. Sometimes the task is even divided among several professionals (e.g., nurse, technician, laboratory staff, or physician).

Target consumers of home medical device and services, on the other hand, are a heterogeneous, primarily nonprofessional group. This may include patients operating the equipment on themselves (self-care) or their proxies—family members (ranging from spouses to grandchildren) and caregivers who operate the equipment for the patient [2,3]. Thus, users' experience, background, ages, level of professionalism, dedication, physical and mental fitness, and language knowledge are variable and unpredictable. The range and breadth of these variables enormously complicate the operation modes and display options of home health care equipment [3]. By and large, target consumers of home health care products and services are people with constraints and disabilities. Their ability to operate the equipment in a home setting may be impaired by various diseases causing motor dysfunction (e.g., Parkinson's disease, essential tremor, or arthritis), visual decline (e.g., diabetes retinopathy, or cataract), the loss of tactile sensation (e.g., diabetes neuropathy), cognitive impairment (dementia), auditory decay (e.g., phonal trauma, presbycusis, or deafness), speech disabilities (e.g., stroke), and equilibrium disorders. Moreover, chronic patients may have multiple illnesses and disabilities and may go through transient and unpredictable changes in their clinical status and performance [8].

The side effects of medications may also disrupt patients' abilities to use health care services and equipment. Such side effects may be changes in visual sensation (e.g., as a result of using anti-cholinergic drugs), auditory abilities (streptomycin), tactile sensation (chemotherapy), alertness level (anti-allergic drugs), perception, cognition and information processing abilities (psychiatric drugs), and more. Common situations such as anxiety, fatigue, loss of sleep and depression may additionally decrease users' attention and performance [2].

Elderly people are a major group of users of home health care devices and services who typically suffer from chronic diseases and a wide array of cognitive and motor disabilities that may significantly disrupt their performance [8].

Given the variety of potential consumer populations, designers and engineers of home health care devices should use human factors and ethnographic methodologies extensively [9–12], applying universal (inclusive) design principles such as equitable use, flexibility in use, simple and intuitive use, perceptible information, tolerance for error, low physical effort, and size and space for approach and use [13,14].

This state of affairs is further complicated as users of health care products and services are not always their purchasers [10]. Health care administrators, insurance companies and regulatory bodies such as the FDA decide what medical devices and technology will be delivered to patients based on social, economic and institutional considerations (e.g., reimbursement) [10].

3. The household environment

Hospitals, clinics and other health care facilities are usually standardized, well regulated, accessible settings operating under close professional supervision and strict regulations. Environmental variables such as sterility (including separation between sterile and non-sterile zones), illumination levels and glare, electromagnetic disturbances, thermal and climate variables, odor and air purity, auditory and noise levels (e.g., regulating alarms and phones), congestion of furniture and visitors are controlled to guarantee appropriate and safe surroundings for equipment performance.

Household and community settings, in direct opposition, are unpredictable and uncontrolled environments. Each user's home is unique and the location of medical devices at home is unknown (by the manufacturer) and changeable, sometimes requiring improvization and compensation when the medical equipment arrives and installed. Electrical interactions with home utensils, a non-sterile environment, cables and line clutter, inappropriate illumination and glare, background noise, dust, distractions and crowding are common situations in household environments. These may restrict and interrupt safe and functional equipment operation and use of health services [3]. When designing home health care products consideration should be given to where and how the equipment is to be set up and used, stored, cleaned, repaired, and disposed of at the end, including not only at home but also at sites such as workplaces, schools, recreation areas, shopping places, and in vehicles while traveling.

The use of telehealth devices and services requires designing not simply the medical item per se but also planning for the entire home setting with all its potential configurations, arranging the location of the wireless sensors and actuators for continuous patient monitoring, setting up a personal and home area network (PAN and HAN, respectively), and creating a secure and confidential environment for sending the information to health care personnel and exchanging clinical data [15]. Even the design of the antenna may have great impact on the quality of the home health care services [16]. Health care devices and services are expected to be integrated into the future 'smart living' [15,17].

In addition to technological and ergonomic issues, health care equipment placed at home is challenged with concerns of aesthetics, design trends, style, fashion and compatibility with the home's interior design, including uncertainness about in which room the equipment will be placed, just as any other consumer product. Furthermore, home health care devices need to avoid the image of sickness or disability [13]. Donald Norman, in his book *Emotional Design* [18], claimed that attractive things work better and that aesthetics and usability correlate. This assumption, which was tested in other fields (e.g., ATMs [19]) should, however, also be investigated with regard to home health care devices.

4. Home health care equipment

The main functions of home health care devices and services are monitoring, documentation, diagnosis, prevention, treatment, alleviation of disease and rehabilitation.

Monitoring and diagnostic devices are available in almost every home, from basic health care equipment such as thermometers and weight scales (e.g., for routine wellness, weight watching or pregnancy follow-up) to specialized devices such as blood pressure monitors, glucometers, INR meters, pulse oximeters, peak flow meters, EEGs, or portable EKG/pulse meters. Most of these monitoring and diagnostic devices are available at pharmacies and usually can be bought over the counter or even via the Internet. At-home documentation of clinical and personal data [personal medical record (PMR)] is developing in parallel, enabling self-follow-up, online connection and consultation with physicians and medical/social staff, and linkage to information sources forming a "disease management team" or, rather, a "wellness management team" [15]. Nevertheless, great effort is needed to adapt patient-oriented interface design and visualization methods to elderly and chronic diseases patients (e.g., Bitterman et al., 2010 [20]).

Home/Self-monitoring and diagnosis capabilities have also expanded due to the innovative technologies of e-health and telemedicine with integrated systems and wireless sensors, allowing multiple and continuous, real time monitoring of patients' physiological signals, plus the documentation and transmission of information [3,21]. Complementary appliances such as accelerometers supply information about patients' daily living activities (posture changes, mobility, and falls) [17,21,22]. Continuous monitoring and follow-up of clinical and physiological signals and daily activities (e-surveillance) can potentially greatly improve the quality of life and wellness of elderly and chronic patients and their ability to stay home safely, with autonomy, dignity and security (smart living) [3,23–25]. Still, researchers have raised some reservations about the benefits of interactive health communication systems for the economic, social and clinical outcomes of the elderly and people with chronic diseases [26–29]. Nevertheless, the consensus seems to be that user-friendly design of health care devices and services is an important criterion for successful home telemonitoring [29].

Home screening using self-diagnosis kits such as pregnancy testing, urinary tract inflammation, sexually transmitted diseases and others may increase the frequency of checking compared to clinic-based testing and decrease screening costs and indirect costs (e.g., time off school or work due to clinic visits, transportation, and childcare) and reduce the number of clinics visits [30].

Disease prevention, *treatment and alleviation devices* extend the responsibility for prevention, further deterioration and treatment from hospitals and clinics to the home and workplace, providing care at the point of need (PON). Home health care devices such as CPAP masks (e.g., sleep apnea), liquid oxygen tanks, oxygen generators, inhalators, catheters, infusion pumps, drug delivery pumps, wound healing equipment, pain relief and analgesia devices (patient-controlled analgesia pump), and even home dialysis machines, however, are relatively complicated systems that may have diverse levels of risk for patients [3] .

Rehabilitation home equipment enables patients to have daily treatments at convenient times using protocols tailored to their specific needs and anthropometrics (personalization). Innovative virtual reality (VR) technology [3] can enrich and upgrade physical activity and increase patient satisfaction, willingness and compliance. Home rehabilitation is in line with the concept of wellness and part of the contemporary trend of gym and fitness equipment that has also moved to the home setting.

Comparison between home- and clinic-based Blood pressure monitoring devices in health care facilities mostly follow the classical model of a sphygmomanometer and stethoscope based on a mercury manometer (considered the gold standard) or more advanced methods of pressure gauge as can be seen in Fig. 1. Professional healthy staff operates the equipment, performing all steps on the patient (placing the cuff around the upper arm, raising the pressure, watching the scale, reducing the pressure, listening to the stethoscope, and documenting and interpreting the data). The blood pressure meter is usually placed on a dedicated platform (at the appropriate height, the same vertical height as the heart or near the patient's bed) ready to be operated.

Fig. 1. Typical blood pressure monitors at health care facilities; classic sphygmomanometers based on inflated cuff around the upper arm and stethoscope supplied with a mercury manometer or a pressure gauge.

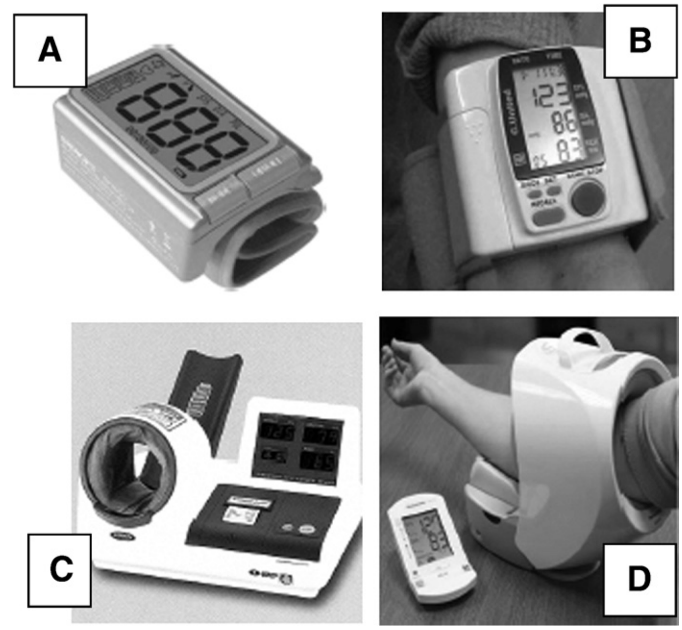

Fig. 2. Various models of blood pressure monitors for home use. A: wrist blood pressure monitor. B: portable arm blood pressure monitor. C and D: fully automated blood pressure measuring machines.

Home blood pressure measurements are performed mostly by non-professional people, either by themselves or assisted by another person. Errors and difficulties may occur at all stages—from deciding or remembering to take the measurement to placing the cuff on the arm, following all procedure stages correctly and precisely, watching and reading the data, remembering the outcome, documenting, understanding and interpreting the implications—so the design must take this into consideration. As can be seen in Fig. 2, various technological efforts have been made to design more inclusive models of monitors to circumvent the difficulties of putting on and adjusting the cuff while having, at the same time, to listen to the pulse and watch the outcome—tasks almost impossible for the non-professional to accomplish separately when self-monitoring, let alone together.

Even a cursory glance at the models in Fig. 2 reveals the problems in their visual display, color selection, bullet size and shape, performance complexity, closeness of elements, and inconsistency in controls and non-intuitive, non-self-explanatory display. A more precise investigation would probably reveal non-compatibility with elderly and chronic patients' disabilities and constraints as presented above. Large variations in monitor configurations and previous exposure to various blood pressure prototypes may result in errors and a danger of applying "negative transfer" in operating the devices. The arrangement of the blood pressure monitoring devices at home (e.g., placed in a too low or high position, disrupting environmental conditions such glare, noise, and vibration) will always be an unknown factor, for which designers must compensate.

5. Summary

Active and continuous involvement of medical personnel, inter-disciplinary experts (e.g., human factors engineers, psychologists, and sociologists) and end users (patients and caregivers) of various ages and genders is needed to redefine the objectives of health care devices and services at home in terms of "all family" use. Principles of inclusive design ("design for all") [2,9] and ethnographic research [12] should be applied, including continuous and long-term follow-up of the devices and services [9,10].

A holistic approach to health care equipment as an integrated part of the overall home surrounding (e.g., smart home) and the emotional experience of wellness is suggested to facilitate personalized services and improved compliance [22].

We believe that if the equipment, the task and the environment are better defined and adapted to the diverse changeable profile of users and their surroundings, errors and accidents will decrease, more patients will become committed users of home services, the function of the collaborative patient–physician disease management team (wellness management team) [15] will be enhanced, and, on the whole, medical and non-medical outcomes will be improved.

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